# Lucas' Theorem II

*Elsewhere (https://www.cut-the-knot.org/arithmetic/combinatorics/LucasTheorem.shtml)* we established the following property of the binomial coefficients:

> *For $p$ prime, integers $a \geq 1$ and $0 < k < p$,*
>
> $$\binom{p^a}{k} \equiv 0 \pmod{p}.$$

This served to derive another useful identity

$$(a+b)^p \equiv a^p + b^p \pmod{p}.$$

Which in turn is used to prove a *theorem (https://www.cut-the-knot.org/triangle/Hanoi.shtml#gen)* due to Édouard Lucas.

> *Let $n, m \in \mathbb{N}$ and let $p$ be a prime. Then*
>
> $$\binom{n}{m} \equiv \binom{\lfloor n/p \rfloor}{\lfloor m/p \rfloor}\binom{n \bmod p}{m \bmod p} \pmod{p}.$$

The theorem can be stated in less formal terms:

> *Let $s, t \in \mathbb{N}$ and let $p$ be a prime. With $0 \leq q, r < p$,*
>
> $$\binom{sp+q}{tp+r} \equiv \binom{s}{t}\binom{q}{r} \pmod{p}.$$

It is worth noting an immediate consequence of Lucas' theorem:

> *Let $n, m \in \mathbb{N}$ and $p$ a prime. Let $q, r$ be the remainders of division by $p$ of $n$ and $m$, respectively. Then $q < r$ implies*
>
> $$\binom{n}{m} \equiv 0 \pmod{p},$$

because $\binom{q}{r} = 0$, for $q < r$. (For, in this case, there is no way to choose $r$ elements out of $q$.) This generalizes the aforementioned property $\binom{p^a}{k} \equiv 0 \pmod{p}$ of the binomial coefficients, and supplies *another example (https://www.cut-the-knot.org/Generalization/epairs.shtml)* of a more general statement being a consequence of a less general one.

## Proof of Lucas' theorem

Let $n = sp + q$, $m = tp + r$,$0 \leq q, r < p$. Then

$$\begin{aligned}(x+1)^{sp+q} &= (x+1)^{sp} \cdot (x+1)^q \\ &\equiv (x^p+1)^s \cdot (x+1)^q \pmod{p}\end{aligned}$$

But then

$$\begin{aligned}(x^p+1)^s \cdot (x+1)^q &= \sum_{i=0}^{s}\binom{s}{i}x^{ip} \cdot \sum_{j=0}^{q}\binom{q}{j}x^j \\ &= \sum_{j=0}^{q}\binom{s}{i}\binom{q}{j}x^{ip+j}.\end{aligned}$$

Now, in the last double sum every power of x appears just once, implying, in particular, that the coefficients by $x^{tp+r}$ on both sides of the equality are the same:

$$\binom{sp+q}{tp+r} \equiv \binom{s}{t}\binom{q}{r},$$

as required.

As an example,

$$\binom{31557}{983} \equiv \binom{2868}{89}\binom{9}{4} \pmod{11}.$$

To compute $\binom{2868}{89}$ we can apply the the theorem again, and so on - if necessary:

$$\binom{31557}{983} \equiv \binom{2868}{89}\binom{9}{4} \equiv \binom{260}{8}\binom{8}{1}\binom{9}{4} \equiv \binom{23}{0}\binom{7}{8}\binom{8}{1}\binom{9}{4} \pmod{11}.$$

And this is immediately $0$ due to the presence of the term $\binom{7}{8}$.

Thus Lucas' theorem, when applied repeatedly, can greatly simplify computing binomial coefficients modulo a prime. This successive of the two numbers $n$ and $m$ by $p$ is nothing but an algorithm for finding their expansions in base $p$. For example,

$$\begin{aligned} 31557 &= ((23 \cdot 11 + 7)11 + 8)11 + 9 \\ &= (((2 \cdot 11 + 1)11 + 7)11 + 8)11 + 9 \\ &= (21789)_{11}. \end{aligned}$$

$$983 = ((8 \cdot 11 + 1)11 + 4 = (814)_{11}.$$

Padding on the left, the example can be rewritten as

$$\binom{31557}{983} \equiv \binom{2}{0}\binom{1}{0}\binom{7}{8}\binom{8}{1}\binom{9}{4} \pmod{11}.$$

In general, we have the following

## Corollary

Let $n, m \in \mathbb{N}$ and $p$ a prime. Assume in base $p$, $n = (n_v n_{v-1} \dots n_1 n_0)_p$ and $m = (m_v m_{v-1} \dots m_1 m_0)_p$, where one of the numbers may need to be padded with $0$s on the left. Then

$$\binom{n}{m} \equiv \binom{n_v}{m_v}\binom{n_{v-1}}{m_{v-1}} \dots \binom{n_1}{m_1}\binom{n_0}{m_0} \pmod{p}.$$

To expand on what we have already observed, if for some $i$, $0 \le i \le v$, $n_i < m_i$, then $\binom{n}{m} \equiv 0 \pmod{p}$. In the binary case, i.e. when $p = 2$, this means that $\binom{n}{m}$ is even whenever $(n)_2$ has digit $0$ in one of the positions where $(m)_2$ has $1$. Compare this with what *we found previously (https://www.cut-the-knot.org/triangle/Hanoi.shtml#lucas)*.

## References


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## Pascal's Triangle and the Binomial Coefficients

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- *Arithmetic in Disguise (https://www.cut-the-knot.org/Curriculum/Geometry/ArithDisguise.shtml)*

- *Construction of Pascal's Triangle (https://www.cut-the-knot.org/Curriculum/Combinatorics/PascalTriangle.shtml)*
- *Dot Patterns, Pascal Triangle and Lucas Theorem (https://www.cut-the-knot.org/Curriculum/Algebra/DotPatterns.shtml)*
- *Integer Iterations on a Circle (https://www.cut-the-knot.org/SimpleGames/IntIter.shtml)*
- *Leibniz and Pascal Triangles (https://www.cut-the-knot.org/Curriculum/Combinatorics/LeibnitzTriangle.shtml)*
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- *Patterns in Pascal's Triangle (https://www.cut-the-knot.org/arithmetic/combinatorics/PascalTriangleProperties.shtml)*
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